
AMBIENT
SYSTEMS
Reimagining the future of smart cities

40%

of global CO2 emissions is attributed to buildings

Climate Impact will largely be felt by building owners and their occupants

$260B+ of portfolio losses
(due to climate change)



Climate Risk can reduce asset value by 20% and annual returns by 40%

Ambitious Net Zero Emission Targets Set by COP UNFCCC (est. 1992) and enforced at national level

‘Conference of the Parties’

United Nations Framework Convention on Climate Change

May 1992-present

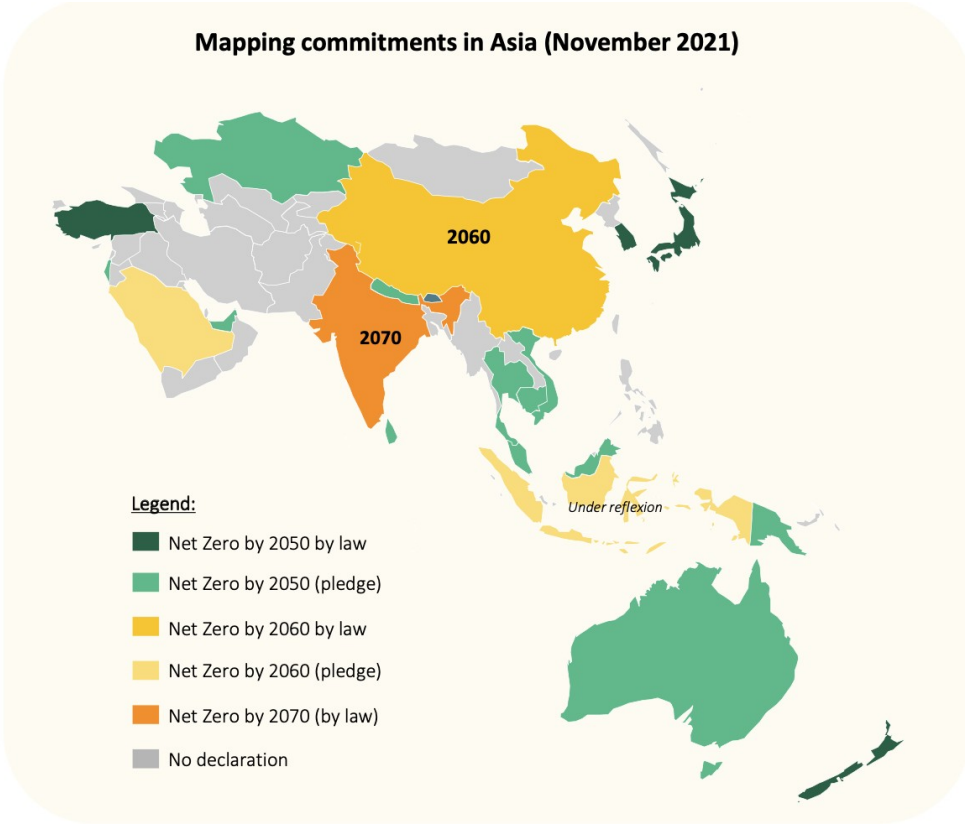
UN CLIMATE CHANGE CONFERENCE UK 2021

IN PARTNERSHIP WITH ITALY

31 Oct 2021 – 12 Nov 2021



13 Nov 2021



Growing list of Carbon Tracking Methodologies

PCAF Partnership for Carbon Accounting Financials

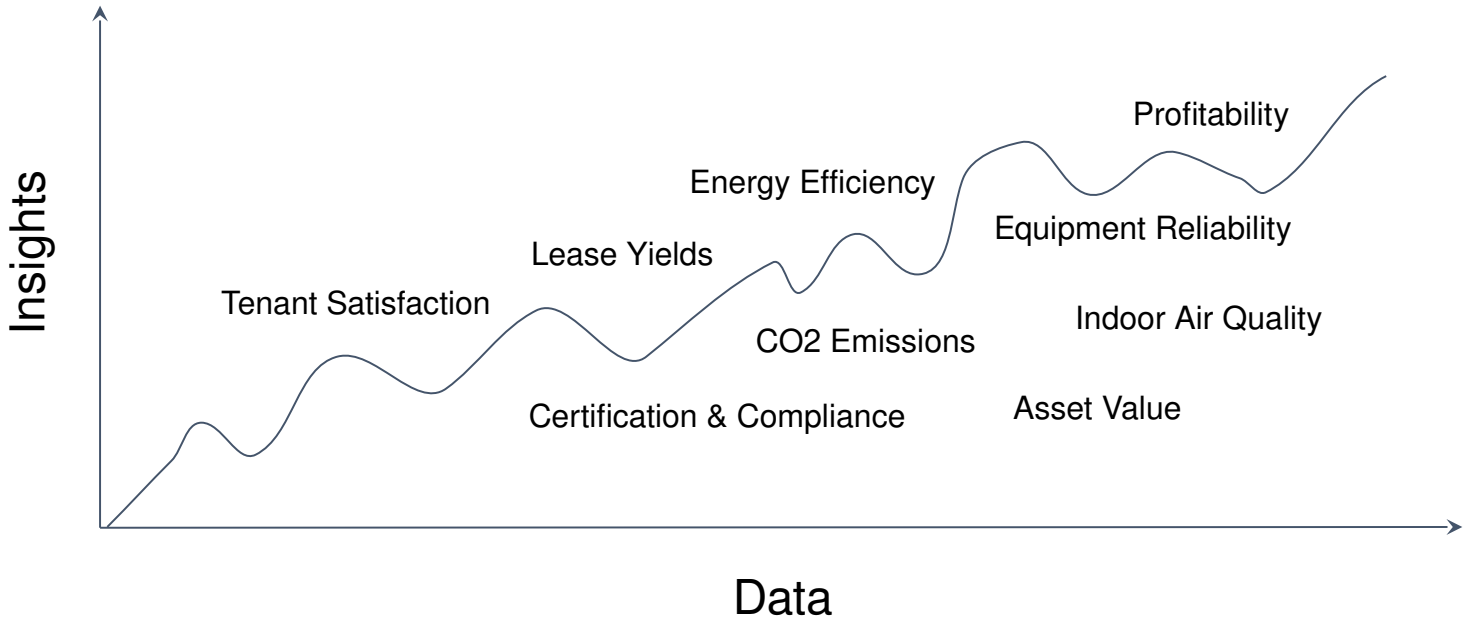
GREENHOUSE GAS PROTOCOL

SCIENCE BASED TARGETS
DRIVING AMBITIOUS CORPORATE CLIMATE ACTION

Global Reporting Initiative™

GRI GSSB Global Sustainability Standards Board

Data Science: empowering real estate asset managers to overcome climate risk



Identify Mispriced Assets
+
Operationalize Climate Resilience
+
Attract ‘Green’ Capital and Tenants

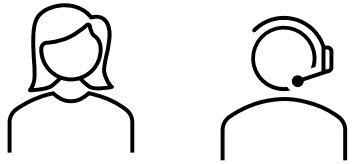
Status Quo

Current Challenges in CRE

Diverse

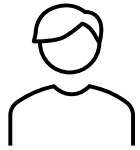


Asset Manager



Facility or Property Manager

Key Stakeholder
Secondary User
Involved in Buying Process
Contributes Data



Occupant (Client)

End Beneficiary
Contributes Data

Gap Between Facility Managers and Property Managers

Facility Management

Driven by technical matters

rsee and manage daily operations
key facility management personnel
estrating maintenance & repair tasks

- Energy management
- Indoor Air Quality
- BMS / BAS operations

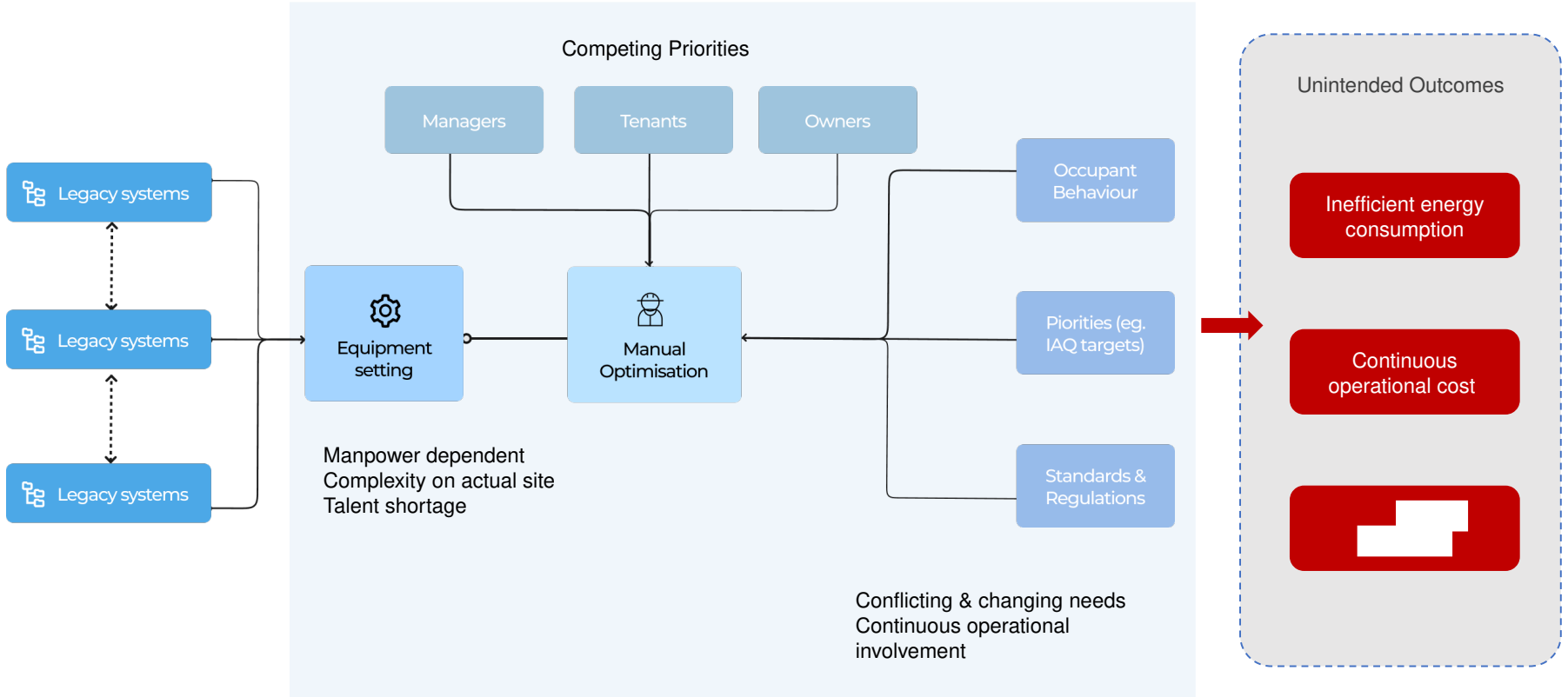
Property Management

Driven by financial matters

ime property value and returns
key personnel including property managers
k with leasing agents

- Communicating with investors and clients (occupiers)
- Limit risks & liabilities
- Ensure lease renewal

Decarbonization & Optimization in Buildings is Complex & Cumbersome



Hard to Find Expertise for Managing Net Zero Smart Buildings

Edge Devices

Automation

User Behavior

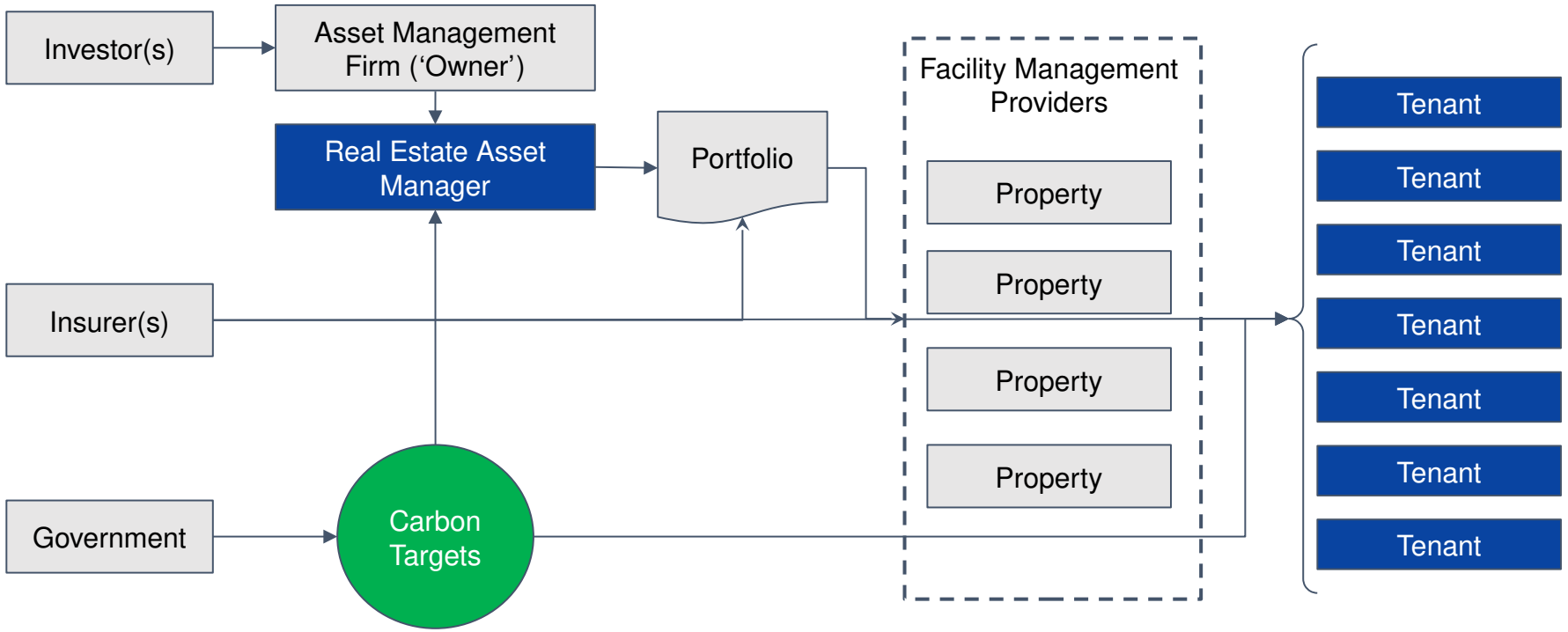


Inefficient energy consumption

Continuous operational cost

High

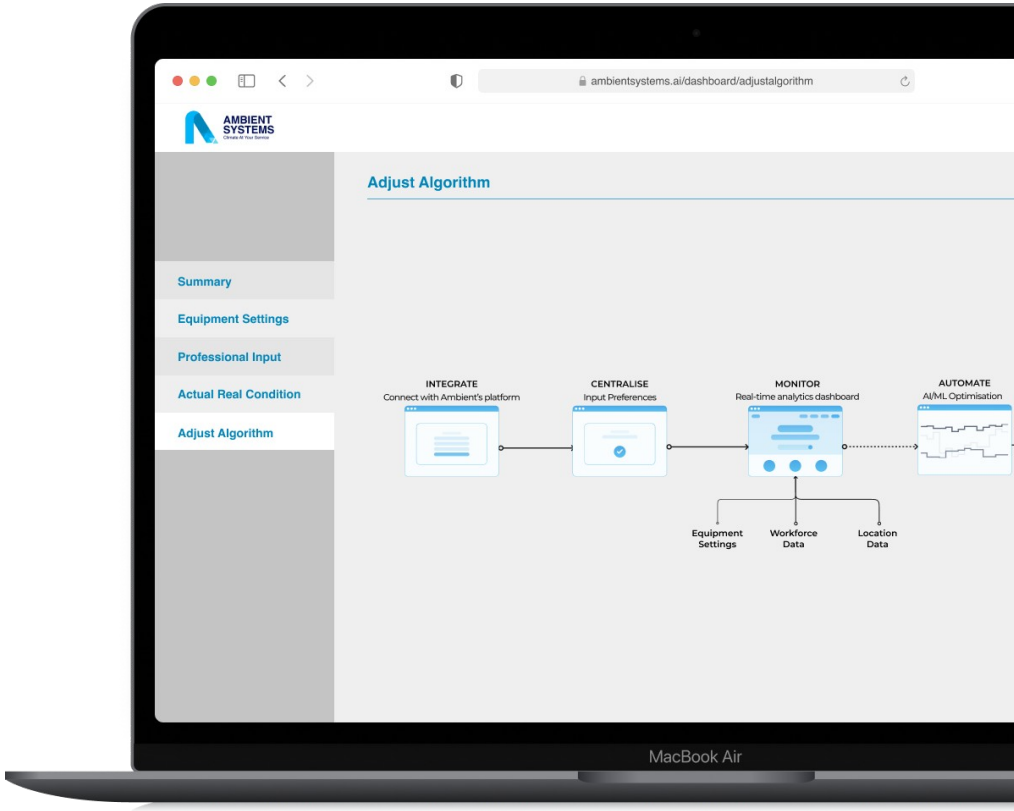
Complex Stakeholder Management Process





Reimagine: Decarbonize your CRE portfolio while driving innovation forward

Decarbonization Intelligence Platform for Commercial Real Estate



Supporting asset managers across the property lifecycle

15% to 25%

Enhanced Valuation Accuracy
(to reflect climate change risk)

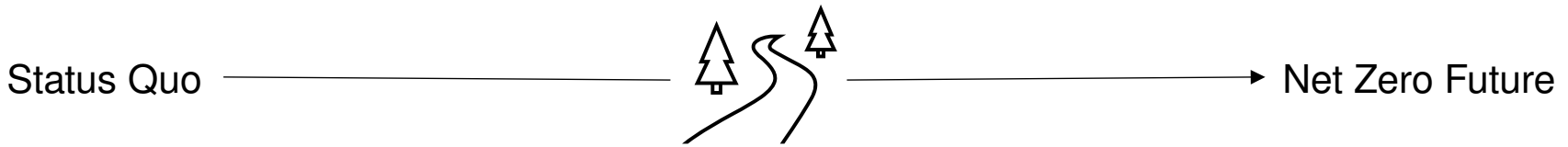
30%

Lower Operating Expense Ratio
(through operational efficiency insights)

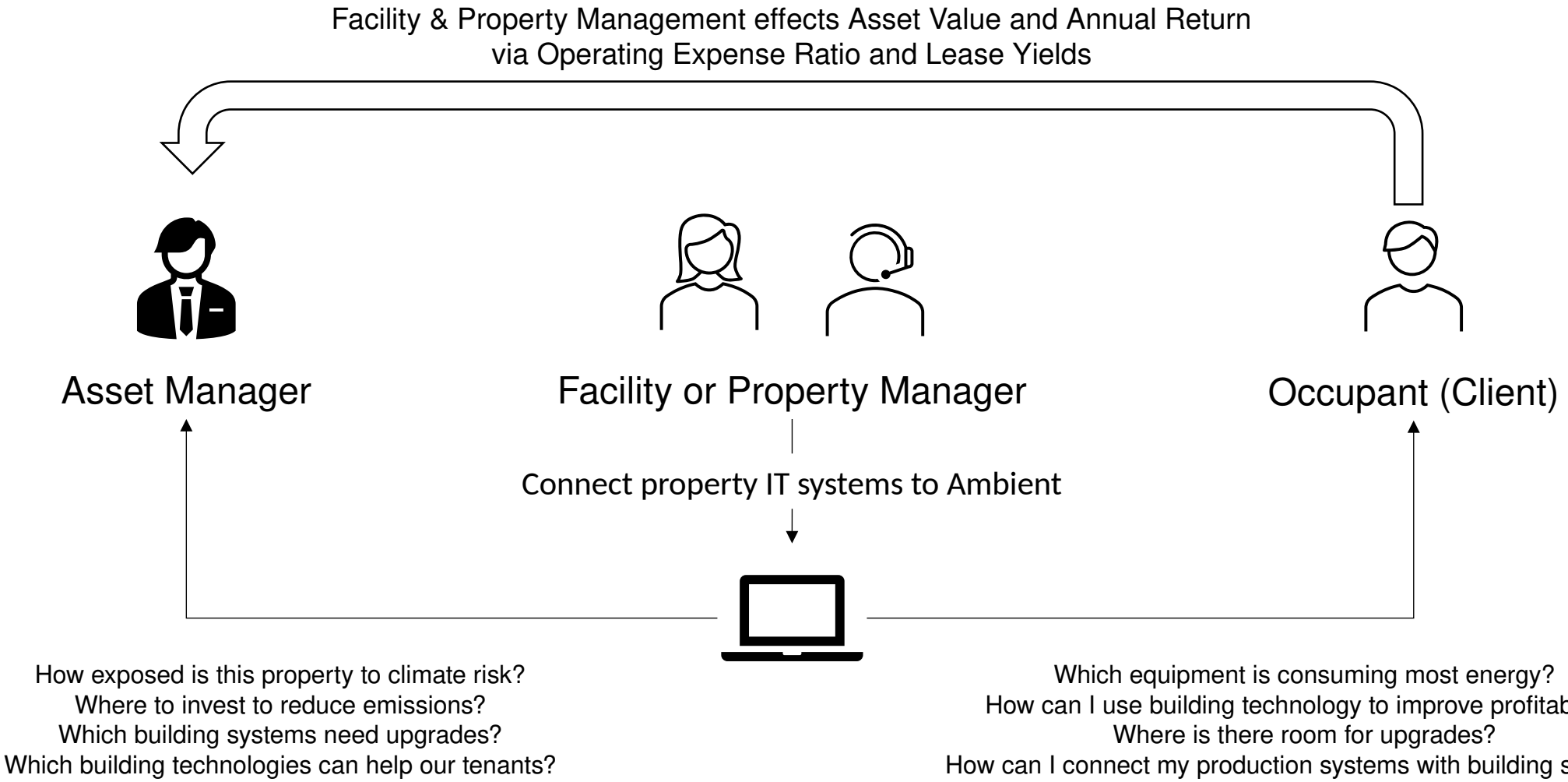
3X

Modeling Time Savings
(compared to current process)

Moving Towards Decarbonization of Portfolio Assets & Occupant Operations



Ambient provides recommendations on how to improve property valuation & profitability



Balanced CRE optimization & decarbonization through three unique features



Flexible On-demand Subscription

No more onboarding, training and manpower reliance. Fast up-time and results through Ambient's subscription service



Simple and Affordable Pricing Plans

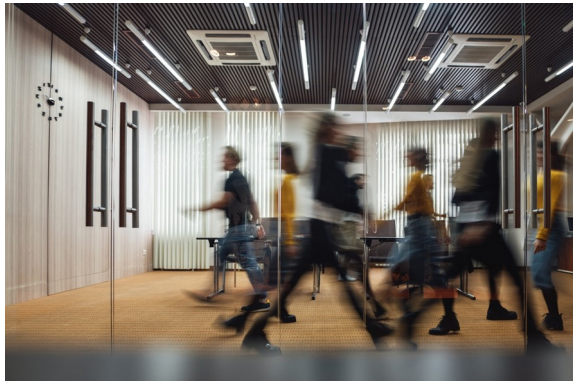
No more reliance on expensive consultants or in-house teams. One platform powered for all your data science use cases



Specialized AI Model Customization

Reach your business goals across energy efficiency, environment, and compliance with Ambient's state of the art AI models

Ambient supports a broad a range of CRE assets

			
Industrial	**Retail**	**Offices**	**Transportation & Logistics**
Manufacturing R&D Life Sciences and BioTech	Malls Shopping Centers Grocery Stores	High Rise Low / Mid Rise Data Centers	Public Transit Airports Warehouses and DCs

Ambient Systems

How it Works

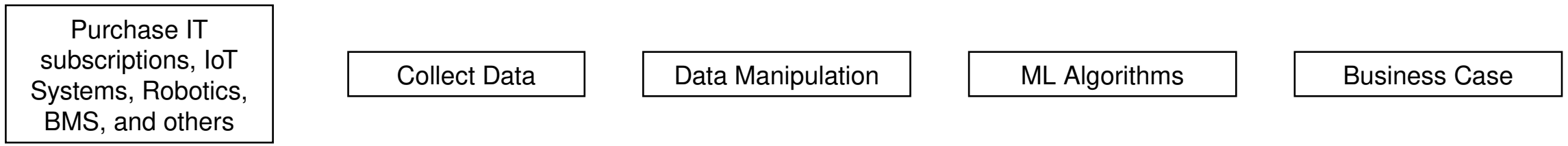
How ML works

ML model can do the same thing as a person, just faster with more accuracy (and often less costly)

ML model cannot infer outcomes without the prerequisite datasets

Sample logical question

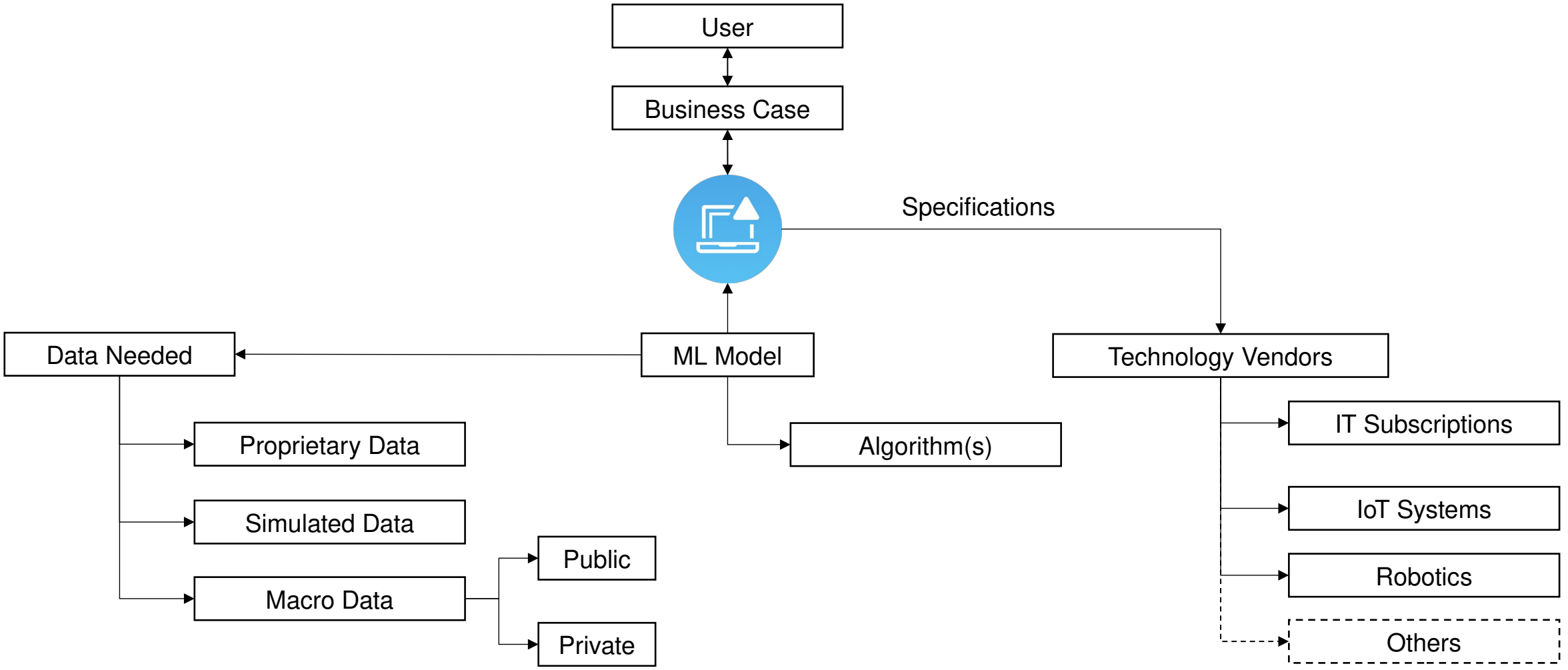
Traditional modelling process is expensive and overlooks key data



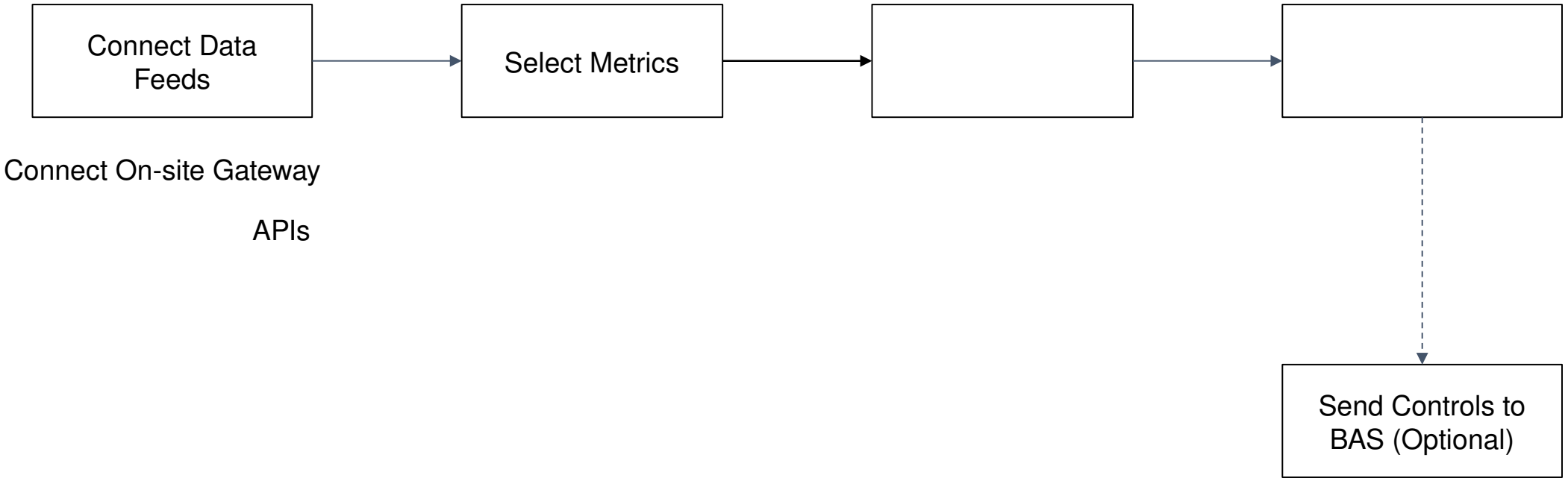
Current modelling process has three data gaps

Incomplete dataset	Mismatching data frequency	Privacy
• Techniques can only mimic human thinking • Cannot deduce outputs without the right inputs	• Half the machine learning algorithms require time series structured, numerical data in large volume • It's easier (and less costly) to collect the data in right intervals than to normalize it down the line	• Many customers do not have all the data (sources) needed to enable machine learning models • Customers do not want their data being shared in other customer's products • Customers also do not want their products being based on other's data

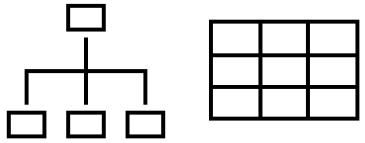
Ambient's modelling process starts with the end in mind



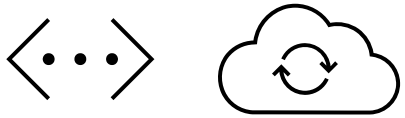
Platform: end-to-end CRE property optimization and decarbonization



Datasets: right datasets are key enablers of the machine learning models



Structured
Time Series Data
GPS Data
System Logs
CRM/ERP



Semi-Structured
Emails
HTML/XML files
JSON files
Network Traffic

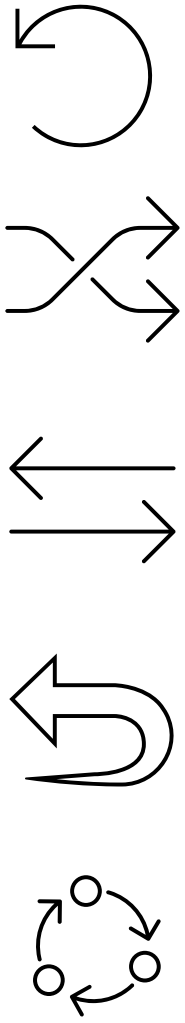


Unstructured
Text
Audio
Video
Images

Variables: Ambient identifies mapping between key inputs and outputs

Independent Variables

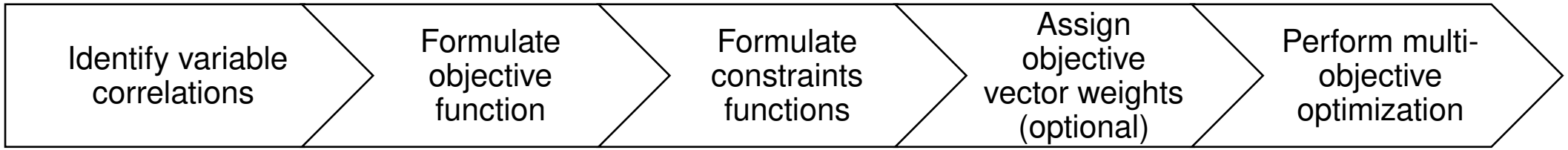
Local Supporting Infrastructure	**New**	**Ambient**
Building Technologies	**New**	
Climate Change Physical Risk	**Climate**	
Climate Change Transition Risk	**Climate**	
Employment Growth in Related Sectors	Conventional	**Baseline**
Employment Rate	Conventional	
Economic GDP Growth	Conventional	
New construction supply	Conventional	
Inflation	Conventional	
Facility Management Costs	Conventional	
Building Construction Type	Conventional	
Property Location	Conventional	

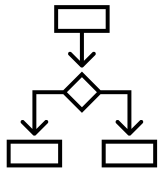


Dependent Variables

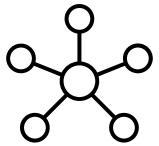
1. Asset Valuation & Appraisal Confidence
2. Occupancy / Vacancy
3. Operating Expense and Operating Expense Ratio

Models & Algorithms: latest machine learning advanced tailored for your portfolio





Expert Systems



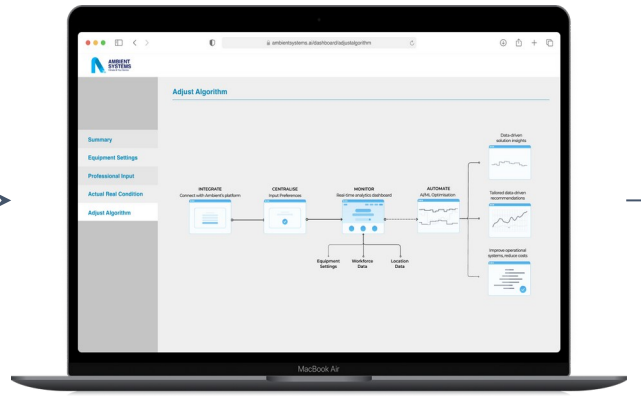
Knowledge Graphs



Stochastic Gradient Descent

Automation: Not a BMS or BAS but Compatible with your and Protocols

Ambient helps your facility
managers make se
data and streamline



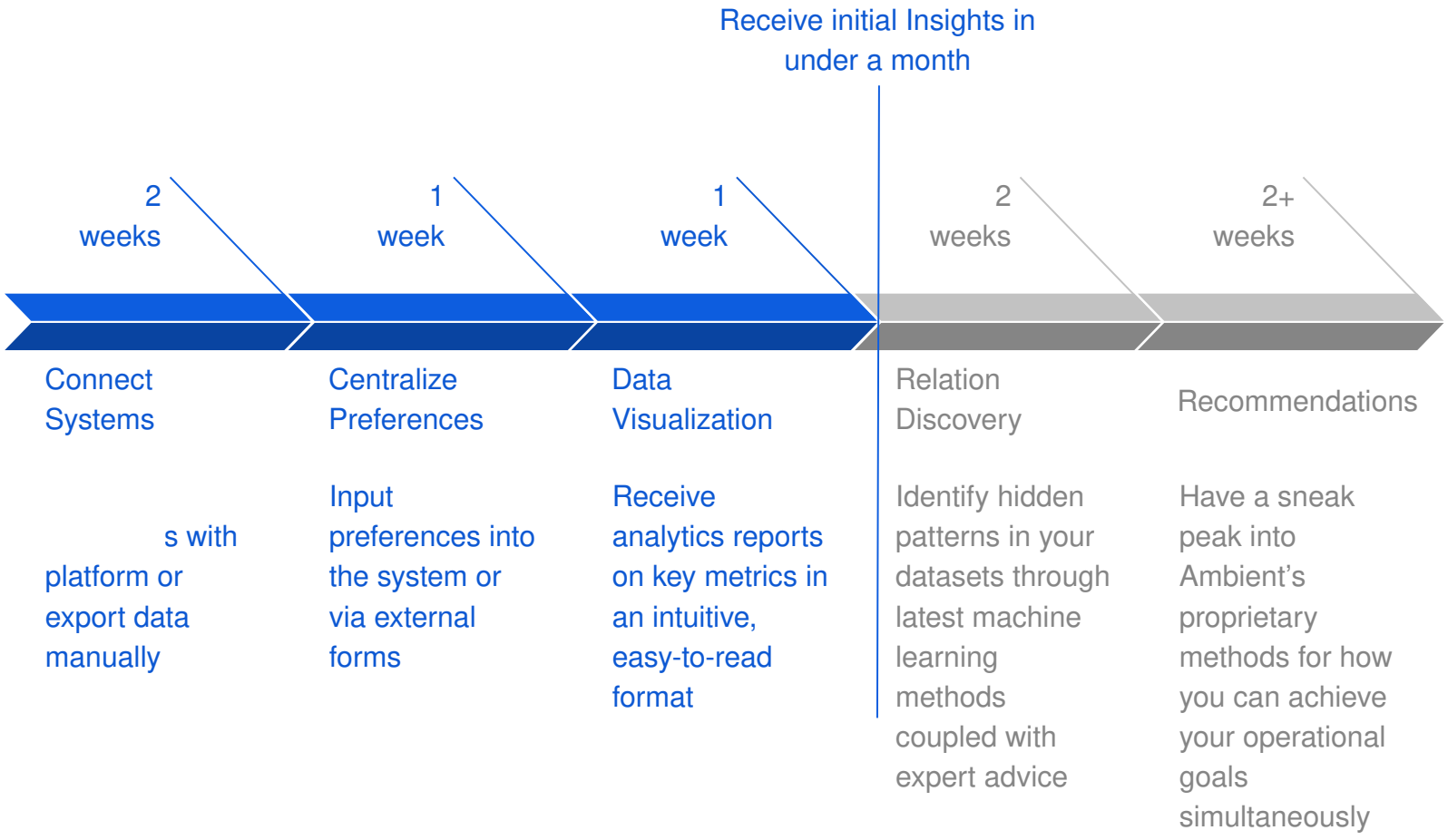
BMS
Database

Siemens Building Technologies,
Sch
, Honeywell, Bosch, ABB,

Moving Forward

POC Project Proposal

POC duration can range 3 to 4 months with initial insights after 2 weeks from setting up the system



Simple steps towards effective POC execution and results

